



**Detect Wildlife → Warn Drivers → Build Biodiversity Intelligence**

**TEAM MEMBERS:**  
Bachanagoni Pranitha  
Lodem Vishnu Vardhan  
Bangla Hasini Reddy  
Danda Sudheer  
Odel Keerthana



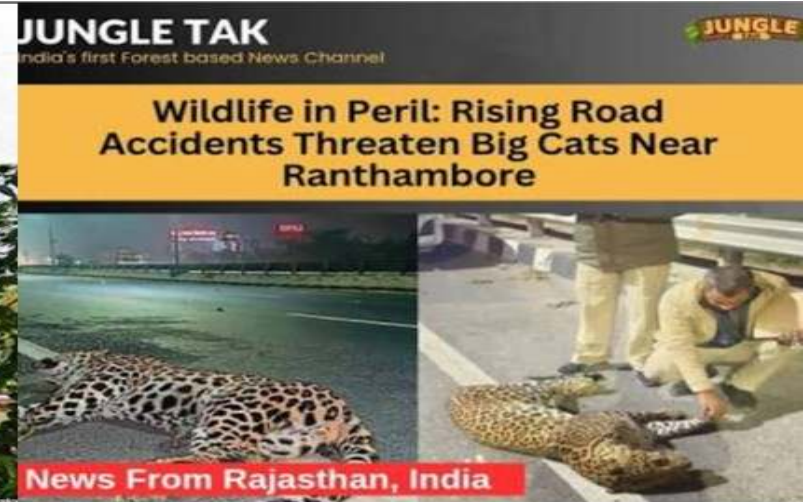


# WILD LIFE + HIGHWAYS

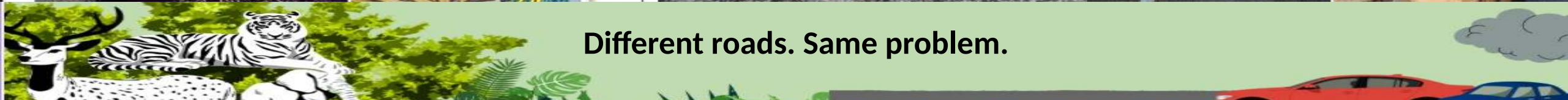
## A DANGEROUS INTERSECTION



**Roar Killed: Tiger fatally knocked down by speeding car on Nanjangud Road**



Different roads. Same problem.

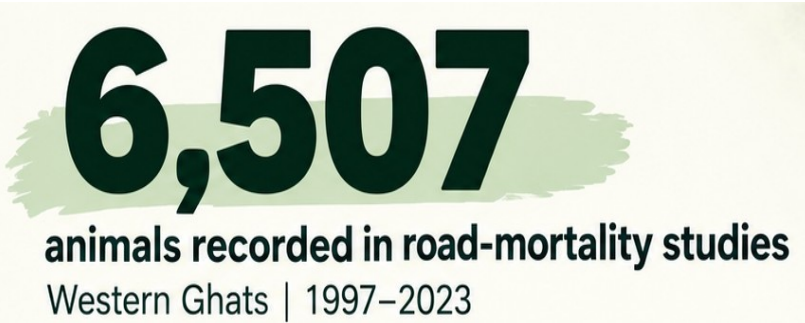
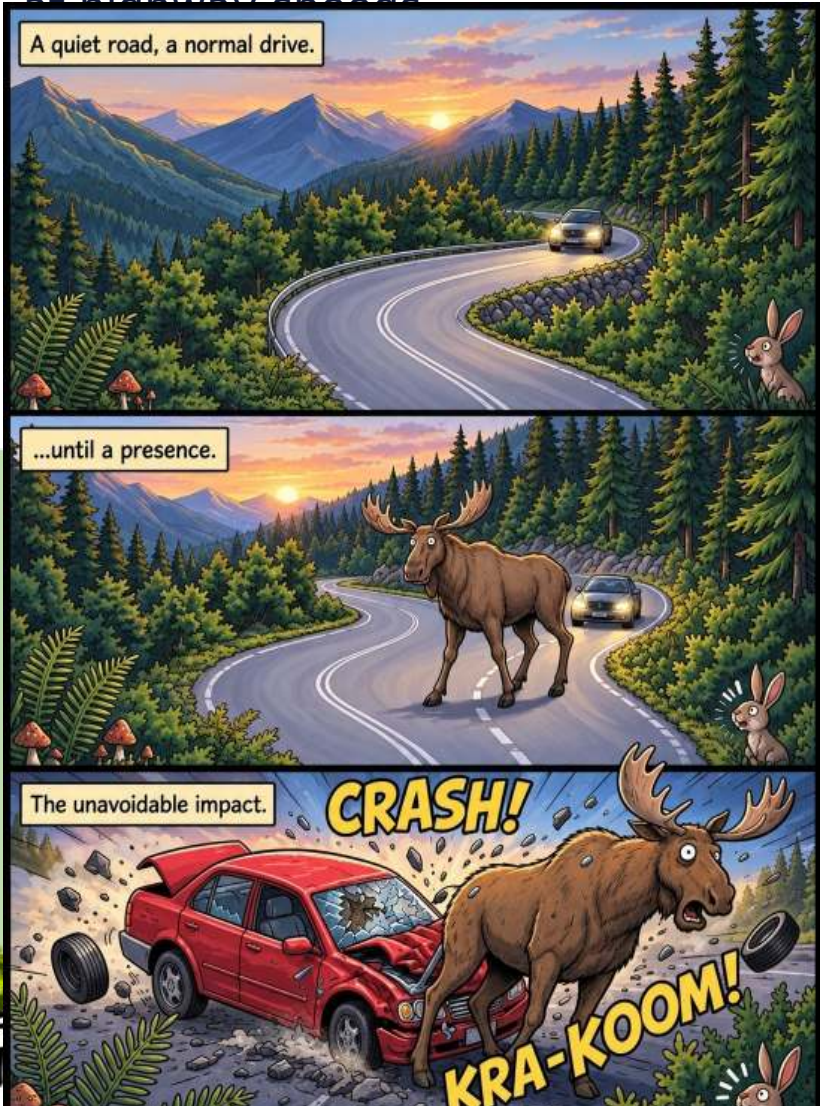




# THE PROBLEM

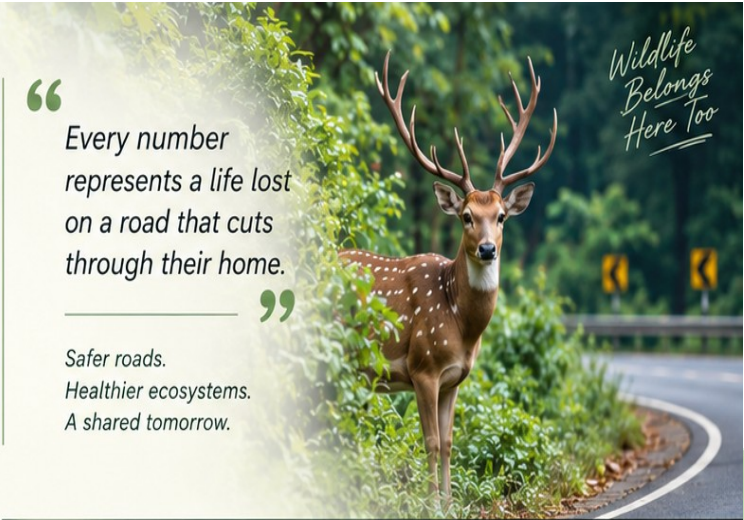
Wildlife movement and high-speed roads often intersect.

Animals move across roads in search of food, water and habitat, while vehicles travel through these corridors at highway speeds.



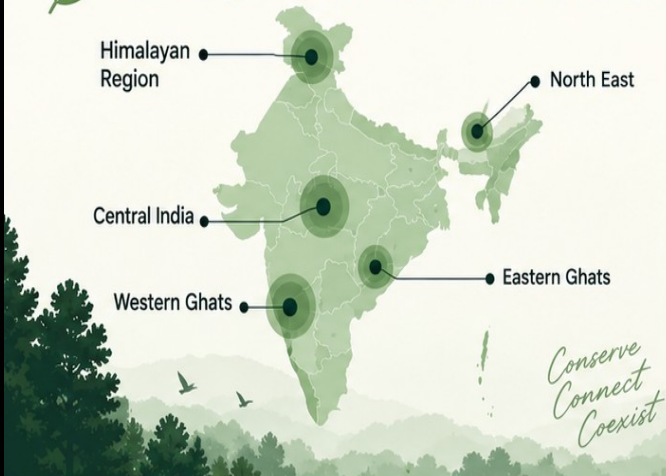
**239 species**  
across multiple  
taxonomic groups

**Represents individual animals**  
recorded/reported as road mortality  
across studies covering 1997–2023.



“  
Every number  
represents a life lost  
on a road that cuts  
through their home.  
”  
Safer roads.  
Healthier ecosystems.  
A shared tomorrow.

## Other reported incidents across Indian road corridors



| Region           | Reported Incidents (Approx.) | Key Species Affected                 |
|------------------|------------------------------|--------------------------------------|
| Western Ghats    | 6,507                        | Elephants, Leopards, Deer, Wild Boar |
| Central India    | 2,000+                       | Tigers, Leopards, Deer               |
| Eastern Ghats    | 1,000+                       | Elephants, Deer                      |
| North East       | 800+                         | Elephants, Hornbills, Other wildlife |
| Himalayan Region | 500+                         | Leopards, Bears, Deer                |

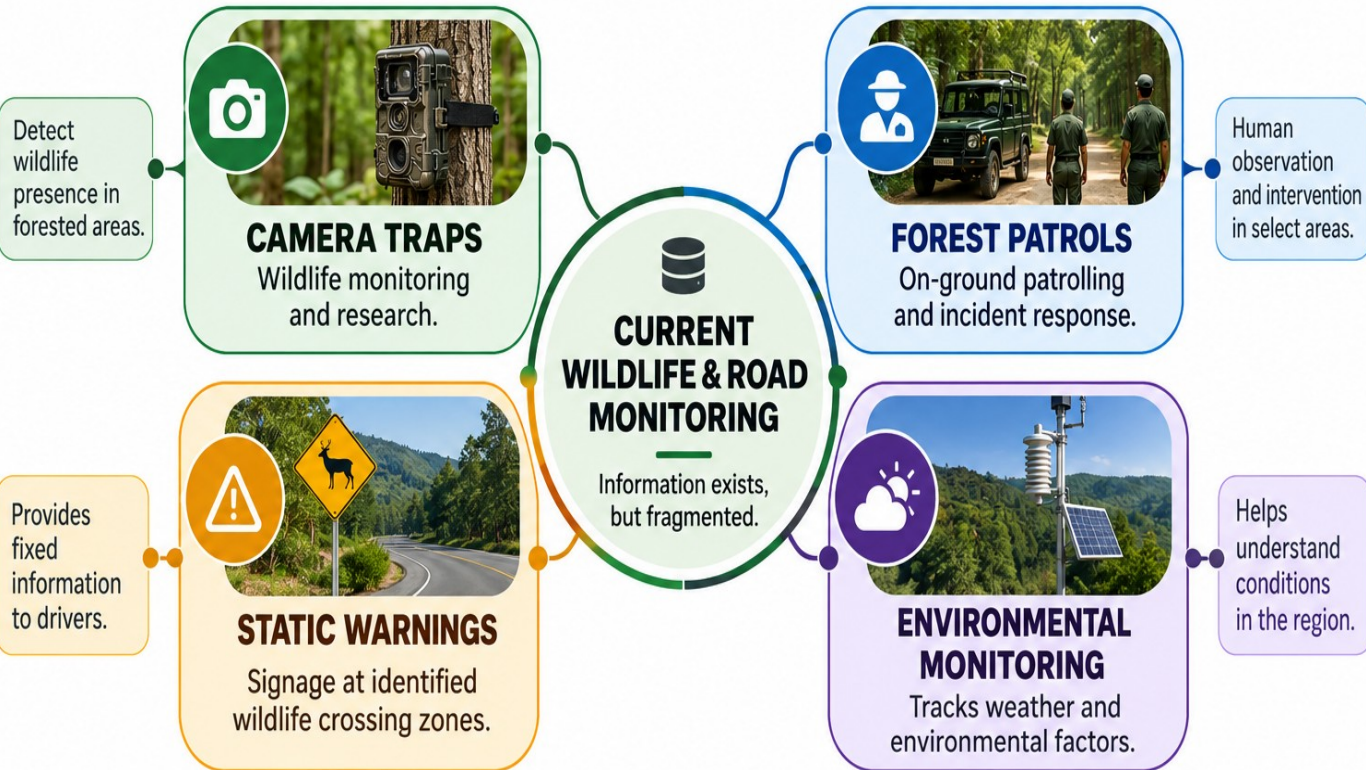


Source: Santhasivam et al., Journal of Wildlife Science, 2026 — Review of wildlife road mortality in the Western Ghats (1997–2023).  
Santhasivam, A. V., Santhanakrishnan, B., Ramachandran, A. P., & Mahaly, M. “Dark side of the development: A Review on Road Mortality of Wildlife in the Western Ghats Biodiversity Hotspot.”  
Journal of Wildlife Science (JWLS), 2026. Published online: 18 February 2026; published: 30 March 2026.



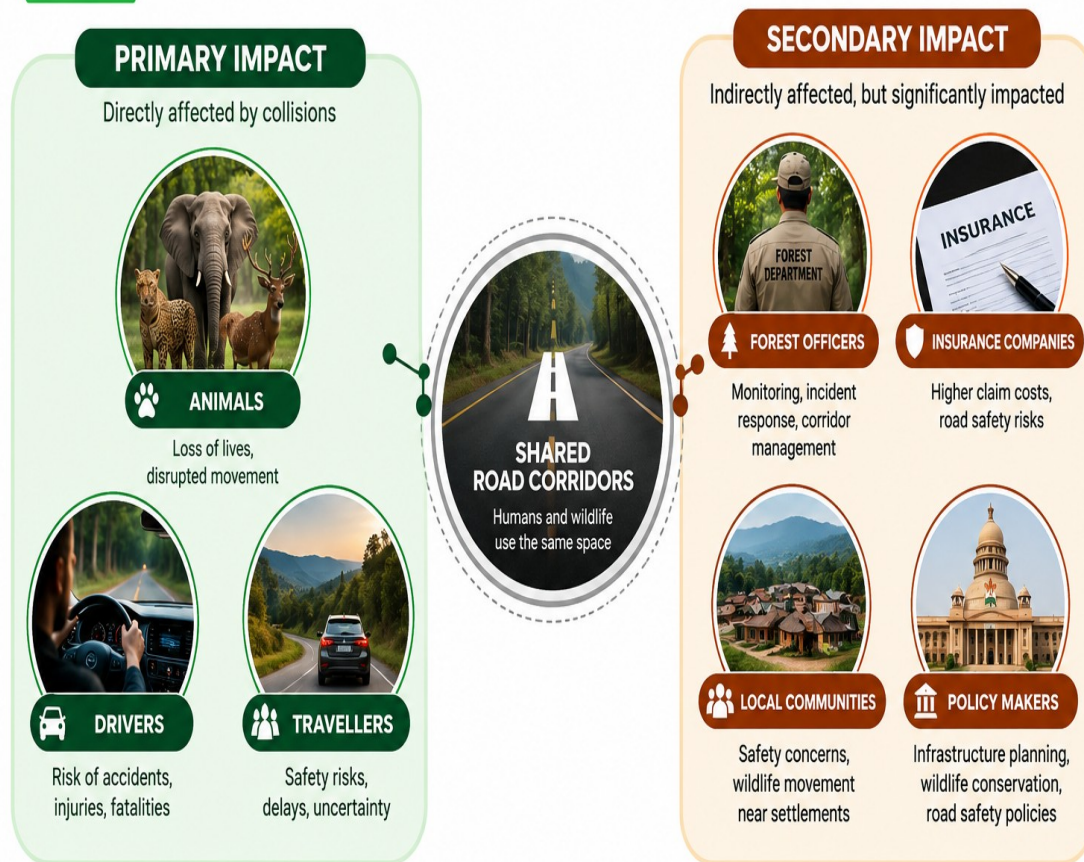
# EXISTING APPROACHES

Multiple efforts exist — but they **work in silos**, not in real time.



These approaches generate valuable information, but they operate independently — **and do not reach the driver when it matters most.**

# WHO IS AFFECTED?



**One road. Multiple stakeholders.**



# OUR SOLUTION

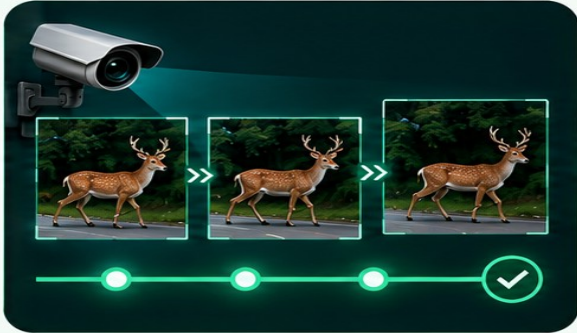
TerraSense is an AI-powered roadside intelligence system that detects and verifies wildlife movement in real time. It connects roadside poles to deliver timely driver warnings while creating actionable data for wildlife conservation.

## 01 DETECT Camera + Edge AI



Detect wildlife locally in real time.

## 02 VERIFY Detection Validation



Confirm the detection across consecutive frames.

## 03 CONNECT Pole-to-Pole Network



Share the alert with configured nearby poles.

## 04 WARN Amber Beacon



Warn approaching drivers before they reach the area.

## ONE SYSTEM. THREE OUTCOMES.

Technology for safer roads and a balanced ecosystem.



### Safer Roads

Reduce wildlife-vehicle collisions.



### Protect Wildlife

Ensure safe passage for animals.



### Smarter Monitoring

Real-time data for better decisions.





# IMPACT

Safer roads. Protected wildlife. A more sustainable tomorrow.



## WILDLIFE PROTECTION

Earlier detection helps prevent wildlife-vehicle collisions and saves lives.



**30-60%**

Potential reduction in wildlife-vehicle collisions



Protection for endangered species



## ROAD SAFETY

Connected warnings give drivers earlier awareness, especially in blind curves and high-risk zones.



**Fewer accidents**

Safer journeys for people and communities



**Increased driver awareness**  
in wildlife corridors



## SUSTAINABLE CORRIDORS

Continuous monitoring builds valuable data to support long-term conservation and better planning.



**Data-driven decisions**

Support smarter conservation planning



**Healthier ecosystems**

Long-term biodiversity intelligence



People. Wildlife. A Shared Tomorrow.



# UNIQUE VALUE PROPOSITION



One detection. Multiple connected warnings. One safer corridor.



## OFFLINE-FIRST

Works without internet dependency.

- ✓ Local communication between poles
- ✓ Reliable in remote areas
- ✓ Continuous operation even in network blackouts



## CONNECTED SAFETY

One detection → alerts nearby poles.

- ✓ Unique local alert system
- ✓ Real-time warnings to drivers
- ✓ Covers extended road corridors
- ✓ Reduces response time



## ENVIRONMENTAL INTELLIGENCE

Every event contributes to a long-term monitoring dashboard.

- ✓ Tracks wildlife movement patterns
- ✓ Monitors environmental conditions
- ✓ Identifies high-risk zones
- ✓ Helps in future planning and conservation

Technology today for wildlife tomorrow





# PILOT ESTIMATE

## KAZIRANGA, ASSAM

 **4 km** wildlife-sensitive road corridor  
Kaziranga National Park, Assam

*Same Roads.  
A Safer Tomorrow.*



### PILOT DETAILS



**LOCATION**  
Kaziranga  
National Park,  
Assam



**CORRIDOR LENGTH**  
**4 km**  
wildlife-sensitive  
road stretch



**NUMBER OF POLES**  
**~15**  
TerraSense poles  
(along the corridor)



**INCLUDED**  
Supply,  
Installation &  
Testing

### COST BREAKDOWN (PER POLE)

| COST BREAKDOWN (PER POLE)                                                            |                                           | PILOT COST<br>(per pole) |
|--------------------------------------------------------------------------------------|-------------------------------------------|--------------------------|
|   | Edge AI board (Raspberry Pi)              | ₹4,000                   |
|   | Camera module                             | ₹1,500                   |
|   | Sensors & communication (PIR, LoRa, etc.) | ₹1,000                   |
|   | Solar panel + battery                     | ₹2,500                   |
|   | Warning system (LED + buzzer)             | ₹1,000                   |
|  | Enclosure + pole + wiring                 | ₹2,500                   |
| <b>TOTAL COST (PER POLE)</b>                                                         |                                           | <b>₹12,500</b>           |



**TOTAL PILOT ESTIMATE**  
(~15 POLES + INSTALLATION & TESTING)

**~₹2.1 LAKH**



PROTECT  
WILDLIFE



SAFER  
ROADS



HEALTHIER  
ECOSYSTEMS

SCALABLE | DATA-DRIVEN | REAL IMPACT





# PROPOSED PILOT METRICS

What we will measure during the pilot deployment.



## Detection Accuracy

Reliability of wildlife detection on the road



## System Uptime

Availability of poles, sensors and communication



## Alert Latency

Time from detection to driver warning



## Wildlife Events Recorded

Number of verified wildlife observations



## False Alerts

Frequency of unnecessary warnings



## Collision Trend

Change in wildlife-vehicle incidents in the pilot stretch

# EXPECTED PILOT OUTCOMES

What we aim to achieve through the pilot deployment.



## Validate the solution

Test wildlife detection and alert system in real-world conditions.



## Evaluate performance

Measure accuracy, response time and system reliability.



## Determine optimal deployment

Identify ideal pole spacing and placement strategy.



## Generate real-world data

Collect and analyze wildlife movement and alert data.



## Assess scalability

Understand requirements for larger deployments.

# OUR CUSTOMERS

Working together for safer roads and healthier ecosystems.

Partners today. Safer tomorrow.



## Forestry Department

Supports forest conservation, forest management and biodiversity protection.



## Wildlife Conservation Agencies

Protects endangered species, prevents poaching and preserves natural habitats.



## NHAI, National Highway Authority of India

Builds and maintains safe, modern and sustainable highway infrastructure.



## Corporate CSR Agencies

Invests in sustainable solutions, supports community well-being and drives environmental impact.



# LIMITATIONS, MITIGATIONS & FUTURE ADVANCEMENTS

Real challenges. Practical solutions. A bigger tomorrow.

## CURRENT LIMITATIONS & MITIGATIONS

| LIMITATION                                                                        |                                                                 |   | MITIGATION                                                                          |                                                                  |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------|---|-------------------------------------------------------------------------------------|------------------------------------------------------------------|
|  | Variable wildlife visibility (e.g., night, camouflaged animals) | → |  | Multi-frame verification and advanced camera placement.          |
|  | False detections in difficult conditions.                       | → |  | Edge-AI filtering and cross-sensor fusion for validation.        |
|  | Connectivity gaps in remote corridors.                          | → |  | Offline-first local node, poirpole communication (LoRa).         |
|  | Limited pilot-scale data.                                       | → |  | Deploy pilot and continuously collect real-world events.         |
|  | Initial deployment cost.                                        | → |  | Optimize hardware and move towards lower-cost, production units. |

## FUTURE ADVANCEMENTS

01



### SMARTER AI

Improve species-level detection and reduce false alerts with larger datasets.

02



### ADAPTIVE WARNING

Use vehicle proximity and speed to make warnings more context-aware.

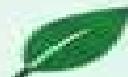
03



### LARGER NETWORKS

Scale from pilot corridors to connected wildlife highways with centralized monitoring.

04



### CONSERVATION INTELLIGENCE

Build long-term wildlife movement and hotspot maps to support conservation planning.



# FIELD INTERACTION & FOREST DEPARTMENT FEEDBACK

Listening. Learning. Building a solution that fits the ground realities.



Interaction with  
DFO & SI  
Achampet Forest Department



*"Insights from the ground make solutions stronger"*

## KEY INSIGHTS FROM THE FOREST DEPARTMENT



WARNING  
MECHANISM



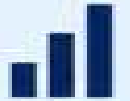
POLE  
PLACEMENT



OFFLINE-FIRST  
DESIGN



CONTROL ROOMS  
& PATROLS



NEED FOR  
THE SOLUTION

## HOW THE FEEDBACK SHAPED TERRASENSE



IDEA



FIELD DISCUSSION



LEARNINGS



DESIGN CHANGES

- Buzzer-based alerts
- General pole placement
- Internet-based system

With DFO & SI,  
Achampet Forest  
Department

- Visual alerts preferred
- Hotspot-based deployment
- Offline-first is beneficial
- Integration with control rooms & patrols is important

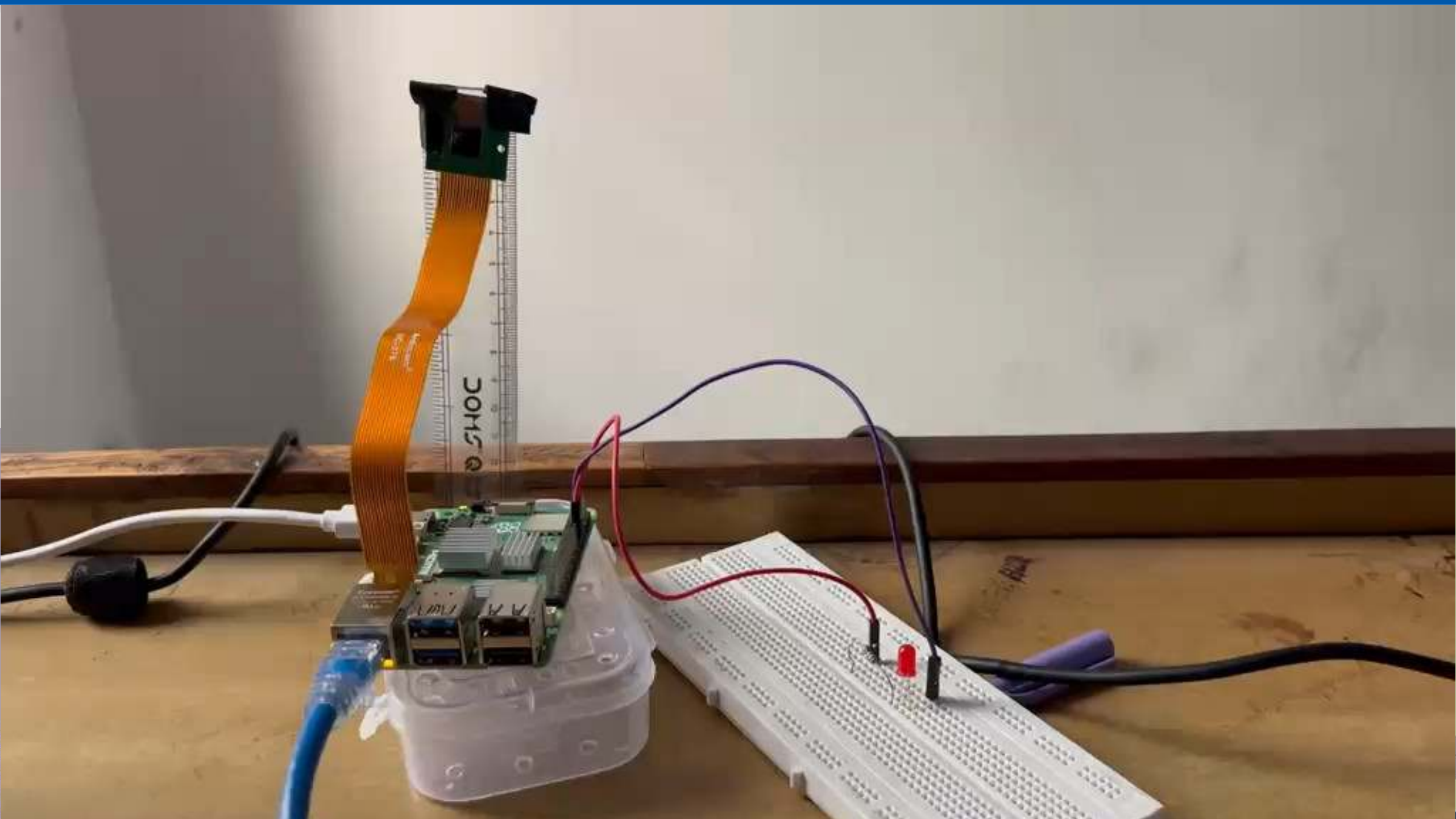
- Amber beacon warning
- Targeted pole placement
- Pole-to-pole (LoRa) network
- Designed to support existing forest operations



Built through collaboration. Ready for real-world impact.









“We are not just monitoring roads;  
**we are protecting the life  
around them.”**



Safer  
Roads



Protected  
Wildlife



Healthier  
Forests



A Safer  
Tomorrow